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Monsoon, 2025

B. Tech 3rd Semester Examination

Building Planning and Computer - Aided Civil Engineering Drawing

Course Code: BCE23303T

Full Marks – 50

Time – 2 hours

*The figure in the margin indicates full marks for the questions.
All questions are compulsory.*

Q1. Multiple Choice Questions (10 × 1 = 10 Marks)

1. The National Building Code (NBC) of India was first published in the year —
a) 1970 b) 1975 c) 1983 d) 1986
2. The plinth area of a building is measured at the —
a) Foundation level b) Plinth level c) Floor level d) Roof level
3. The National Building Code (NBC) is published by which organization?
a) ISI b) BIS c) CPWD d) HUDCO
4. A Green Building primarily focuses on —
a) Aesthetic appearance b) Energy efficiency and sustainability c) Cost reduction only
d) Luxury interior design
5. The primary function of a soak pit is to —
a) Store rainwater b) Treat solid waste c) Dispose of wastewater d) Pump groundwater
6. The wet riser system in a building is used for —
a) Electrical supply b) Firefighting c) Drainage d) Air conditioning
7. The lighting design of a staircase should ensure —
a) Decoration b) Minimum illumination c) Safe movement d) Energy wastage
8. The main power supply to a building enters through —
a) Distribution board b) Meter box c) Substation d) Switchboard
9. The carpet area of a building includes —
a) Walls b) Balcony c) Veranda d) Room floor area
10. The main function of a sump in a water supply system is —
a) Filtration of water b) Collection and storage of water c) Pumping of sewage d) Water treatment

Q2. a) Explain the principles of building planning according to the National Building Code (NBC). Discuss how site selection and orientation influence the efficiency and comfort of a residential building. (10)

OR

Q2. b) Explain the execution of residential and public building plan. Include the role of the National Building Code and the concept of built environment in planning. (10)

Q3. a) Compare and contrast water supply and drainage systems in residential and public buildings. Explain how solid waste management systems can be integrated into modern building design. (10)

OR

Q3. b) Identify the factors affecting building planning. Justify how function, topography, climate, building by-laws and orientation influence building design. (10)

Q4. a) What are the basic components of domestic electrical supply and types of wiring systems. Write the use of solar panels and automatic water level controllers in modern buildings. (10)

OR

Q4. b) What are the fire protection system in buildings with emphasis on: (10)
i) Wet and dry risers
ii) Sprinkler systems

Q5.a) Design a staircase for a three-storey general hospital building ensuring compliance with safety, accessibility, and functional norms. Include necessary calculations and justification for your design choices. Each storey floor-to-floor height is 3.6 m. The stair hall allocated on each floor measures 3.0 m × 6.5 m (width × length). The stair must allow stretcher movement and emergency evacuation. (10)

OR

Q5.b) Design a staircase for a three-storey boutique clinic (not a hospital) where space is limited. Each storey floor-to-floor height is 3.3 m. The stairs must serve normal circulation (no stretchers) and comply with accessibility and egress safety rules for public use (but it will not be used for patient emergency transfer). Assume the missing data. (10)